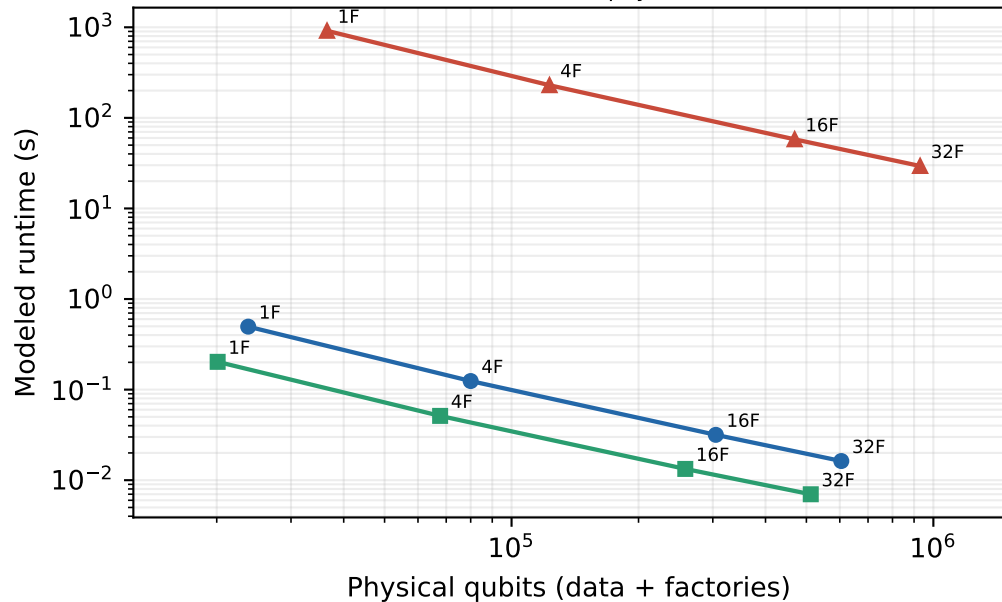


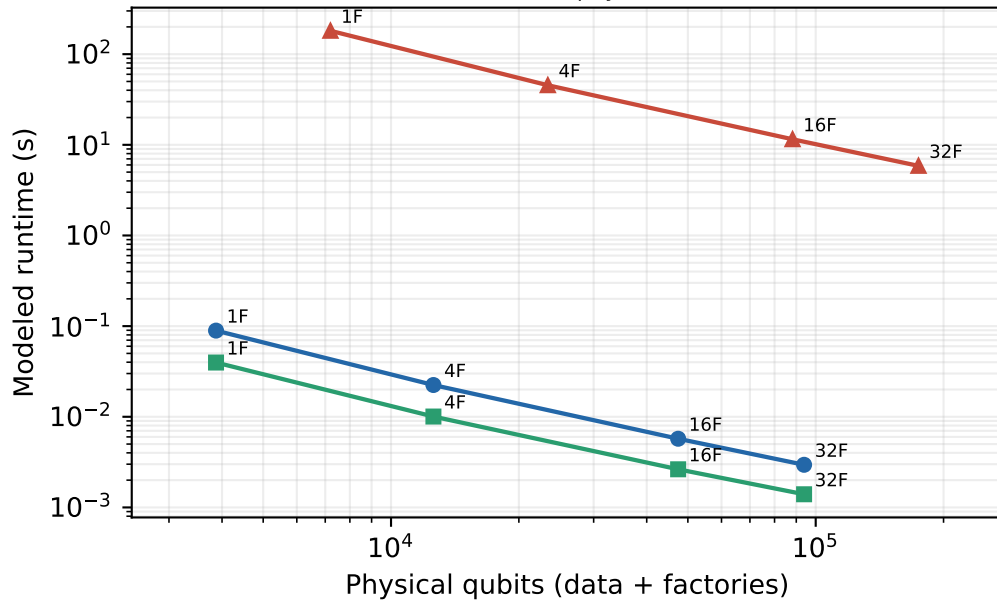
Measured compiler output $\rightarrow$ actual staq synthesis $\rightarrow$ QRE Pareto grid

20k witness, $\varepsilon_{total} = 10^{-6}$, equal synthesis allocation

Conservative: $p_{phys} = 10^{-3}$



Improved: $p_{phys} = 10^{-4}$



—●— Semantic-first (UCC)

—■— External semantic (TKET)

—▲— Materialize-first (Qiskit)